

Myosin Motor Model Visual Atlas

Revision 2 - niche comparators and novelty-claim audit

Added in this revision

- MUSICO: 3-D sarcomere lattice and thousands of spatial head-site interactions
- FiberSim: individual heads, dimer pairs, compliant thick/thin filaments
- Vilfan gliding/twirling: helical actin geometry and filament rotation
- Elastic-lever myosin V: two-head structural mechanochemistry
- Spatial half-sarcomere models: radial/axial geometry controls recruitment
- Fluctuating-ratchet actomyosin: continuous stochastic energy landscapes
- Motor-to-adhesion traction model: nanoscale-to-microscale coupling

Central finding: SoftBox's defensible novelty is the combination of features, not an absolute maximum in system DOF or detail.

All drawings are original simplified reconstructions.

Novelty claim audit

Unsafe absolute claims

- most detailed myosin model
- largest number of degrees of freedom
- first 3-D mechanochemical model
- first explicit individual-head model
- first explicit two-head geometry
- first nanoscale-to-microscale bridge

Defensible positioning

SoftBox occupies an uncommon intersection: explicit myosin-II internal mechanics (head/converter/lever/F8/S2/HMM fork), state-resolved chemistry, spatial binding to a flexible Brownian actin filament, and gliding/single-molecule validation in one framework.

Closest threats to broad claims

MUSICO and FiberSim (system/head scale); Vilfan myosin V (two-head mechanics); Vilfan twirling (gliding geometry).

Therefore use a qualified combination claim, not a priority claim based on raw DOF.

Feature matrix

model	heads	2-head	actin sites helix	head/lever mechanics	S2/tail mechanics	flexible Brownian actin	detailed chemistry	large gliding ensemble
Huxley 1957 cross-bridge model	○	○	○	○	○	○	○	○
Huxley-Simmons 1971	○	○	○	●	○	○	○	○
Duke 1999 molecular model	○	○	○	●	○	○	●	○
Walcott-Warshaw-Debold 2012	●	○	○	○	○	○	●	●
Parallel Cluster Model	○	○	○	○	○	○	○	○
Bipolar minifilament tug-of-war	○	○	○	○	○	○	○	○
Jung et al. 2020 motility-assay model	●	○	●	○	○	●	○	●
Cytosim / Nédélec-Foethke	●	○	○	○	○	●	○	○
aLENS / explicit crosslinking motors	●	●	○	○	○	○	○	○
Ding et al. explicit myosin-II filament architec	●	○	●	○	○	○	○	○

Filled circle = explicitly represented or central to model; open circle = absent or strongly reduced.

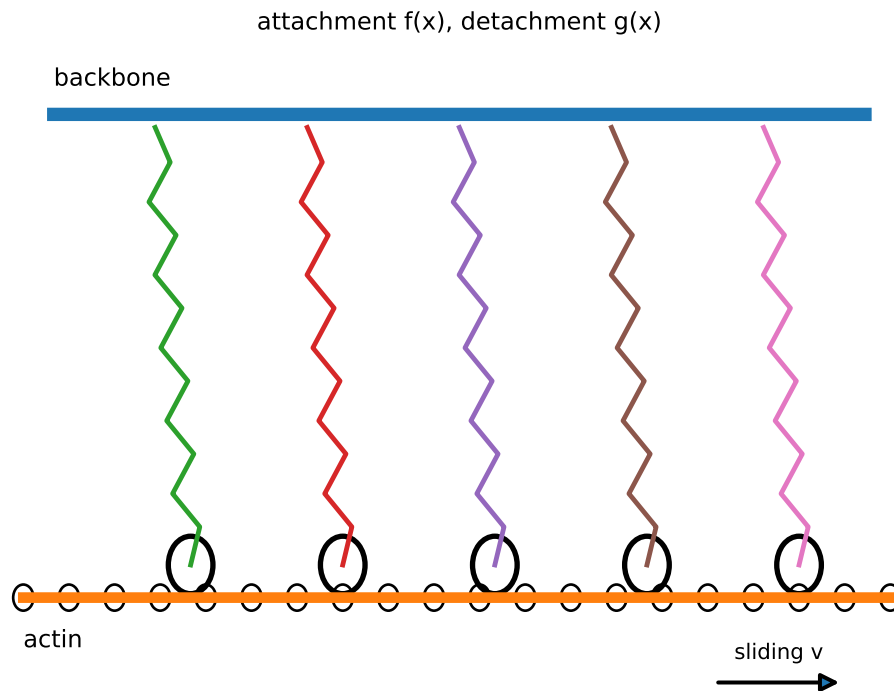
Feature matrix

model	heads	2-head	actin sites helix	head/lever mechanics	S2/tail mechanics	flexible Brownian actin	detailed chemistry	large gliding ensemble
Adamović-Mijailovich-Karplus S2 mechanics	○	○	○	○	●	○	○	○
SoftBox canonical motor	●	●	●	●	●	●	●	○
MUSICO 3-D sarcomere lattice	●	○	●	○	○	○	●	●
FiberSim half-sarcomere	●	●	●	○	○	○	●	○
Vilfan 2009 gliding/twirling model	●	○	●	○	○	○	○	○
Vilfan 2005 elastic lever-arm myosin V	●	●	●	●	○	○	●	○
Spatially explicit half-sarcomere lattice model	●	○	●	○	○	○	○	○
Chemically driven fluctuating-ratchet actomyosin	○	○	○	○	○	○	○	○
Cell-traction multiscale individual-head model	●	○	○	○	○	○	○	○

Filled circle = explicitly represented or central to model; open circle = absent or strongly reduced.

1. Huxley 1957 cross-bridge model (1957)

Classical distributed cross-bridge



Primary citation

A. F. Huxley, Muscle Structure and Theories of Contraction, Prog. Biophys. Biophys. Chem. 7, 255–318 (1957).

Atlas figure: original simplified reconstruction.

Motor representation

Many identical elastic side-pieces attached to a rigid myosin backbone; actin slides past them.

States

Detached / attached, with attachment and detachment rates depending on cross-bridge extension x .

Defining laws

Transport-reaction equation for attached-head density $n(x,t)$; force is the integral of elastic strain over attached cross-bridges.

Spatial scale

1-D strain coordinate; filament-level continuum population.

Stochasticity

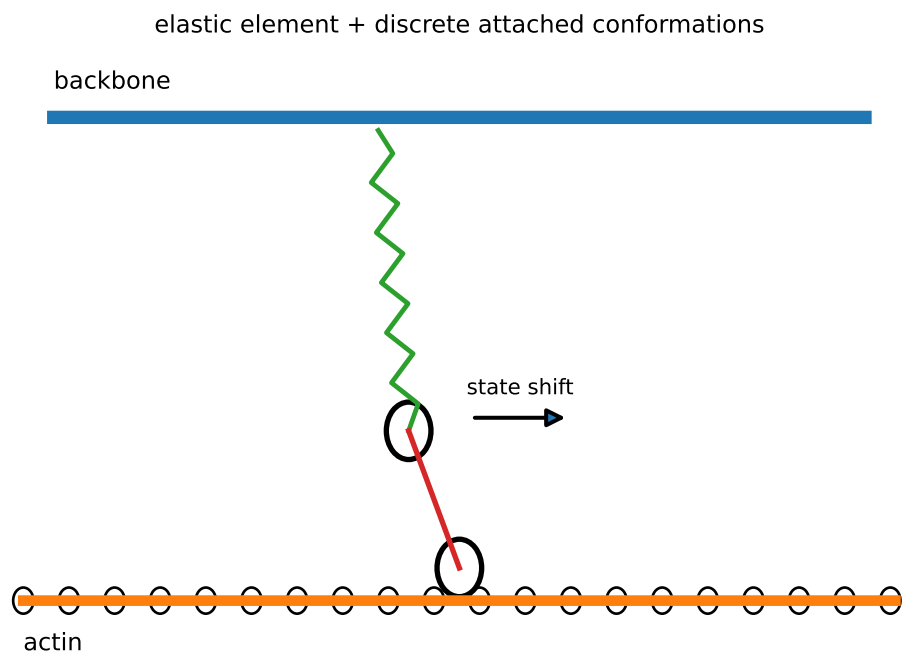
Rates are probabilistic at the head level, represented as population kinetics.

Relevance to SoftBox

SoftBox resolves individual motors, Brownian filament motion, chemistry, head geometry, and S2 mechanics rather than a strain-density field.

2. Huxley-Simmons 1971 (1971)

Discrete conformational cross-bridge



Primary citation

A. F. Huxley and R. M. Simmons, Nature 233, 533–538 (1971), doi:10.1038/233533a0.

Motor representation

An attached elastic cross-bridge with multiple discrete conformational states.

States

Attached conformations separated by rapid reversible transitions; each conformation has a different preferred extension.

Defining laws

State-dependent elastic free energies; length steps redistribute state occupancy and produce fast force recovery.

Spatial scale

Single cross-bridge coordinate embedded in a muscle ensemble.

Stochasticity

Thermal transitions among attached conformations.

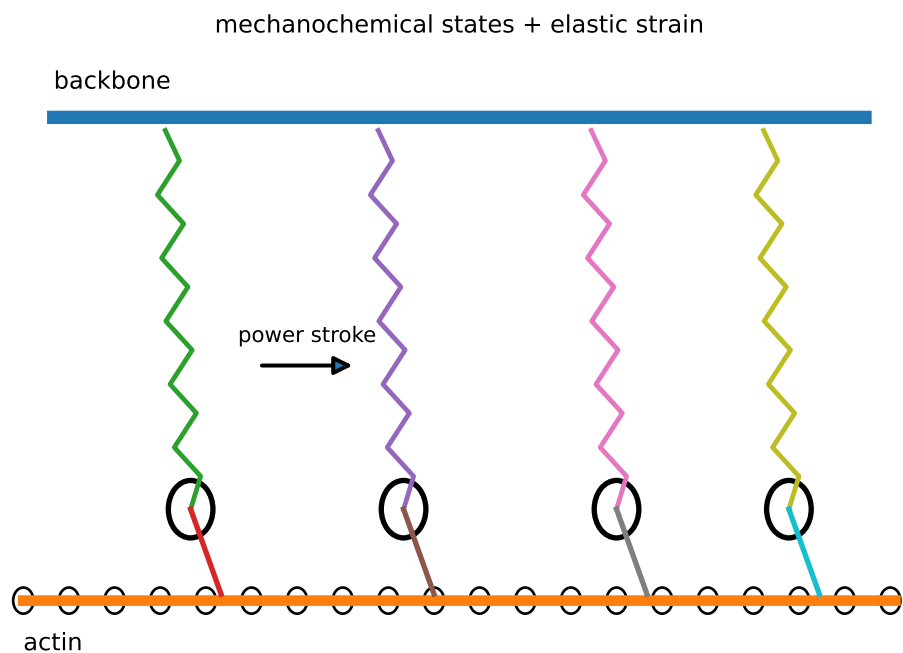
Relevance to SoftBox

Closest historical ancestor of SoftBox’s explicit power-stroke geometry, but lacks explicit 3-D head, S2, binding geometry, and nucleotide cycle.

Atlas figure: original simplified reconstruction.

3. Duke 1999 molecular model (1999)

Mechanochemical ensemble



Primary citation

T. A. J. Duke, Molecular model of muscle contraction, PNAS 96, 2770–2775 (1999), doi:10.1073/pnas.96.6.2770.

Motor representation

Individual cross-bridges cycle through biochemical/mechanical states and interact through common filament motion.

States

Detached, pre-power-stroke, post-power-stroke, and nucleotide-dependent transitions.

Defining laws

State transition rates obey mechanochemical energetics; each attached head has elastic strain and a state-dependent stroke.

Spatial scale

1-D sarcomere / common sliding coordinate.

Stochasticity

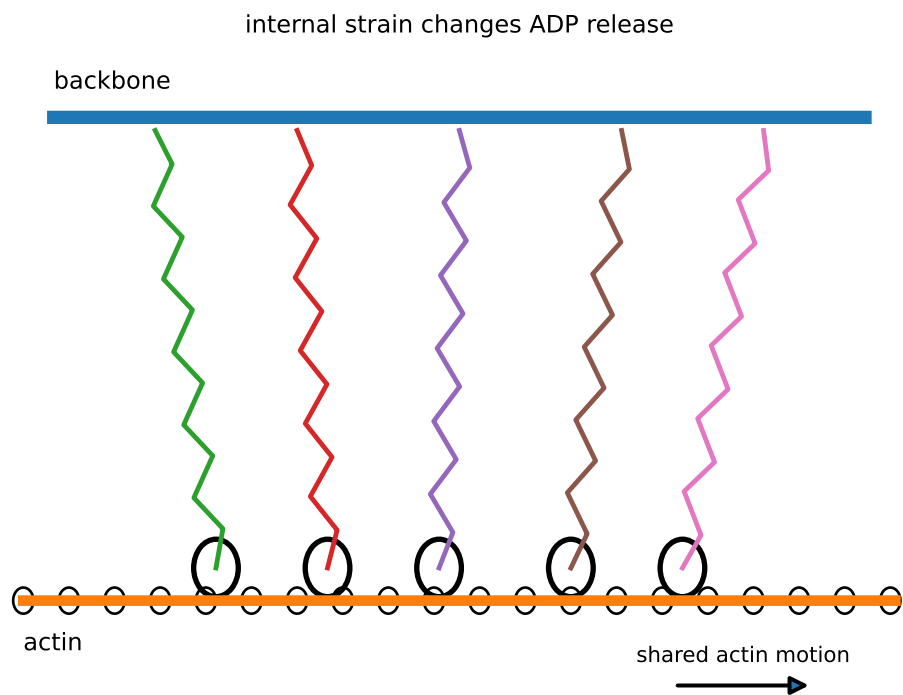
Monte Carlo state transitions for many motors.

Relevance to SoftBox

Very close in mechanochemical philosophy; SoftBox adds explicit head/lever/S2 geometry, surface binding, flexible actin, and assay-level Brownian dynamics.

4. Walcott-Warshaw-Debold 2012 (2012)

Mechanically coupled myosin ensemble



Primary citation

S. Walcott, D. M. Warshaw, E. P. Debold, Biophys. J. 103, 501–510 (2012), doi:10.1016/j.bpj.2012.06.031.

Motor representation

Multiple elastic myosins share one actin coordinate; internal forces alter ADP-release kinetics.

States

Weak/unbound and strongly bound states with ADP release followed by ATP-induced detachment.

Defining laws

Elastic force from head strain; force-dependent ADP release; common actin displacement mechanically couples otherwise independent motors.

Spatial scale

1-D motility assay / ensemble coordinate.

Stochasticity

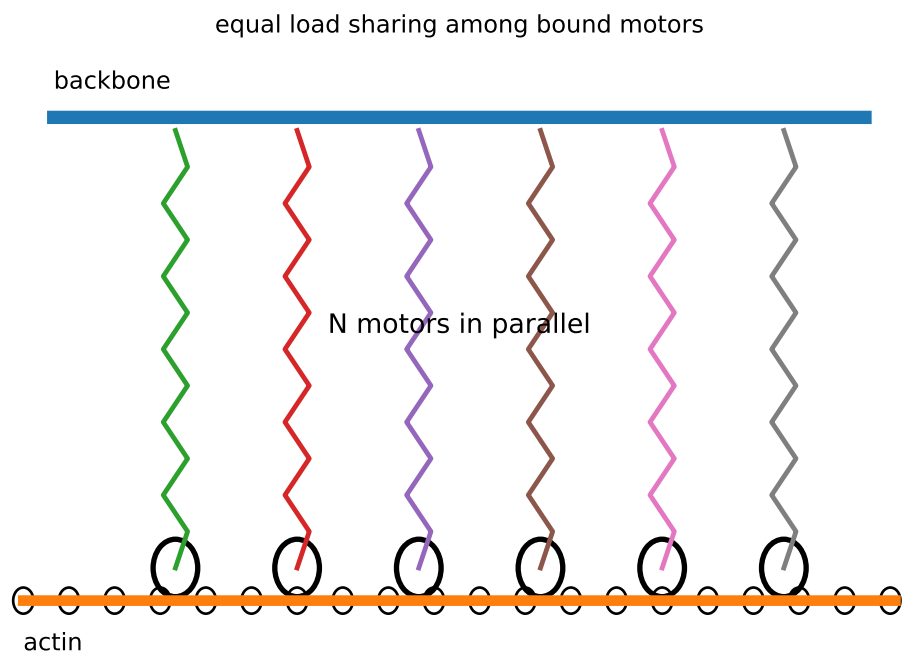
Individual binding and chemical events simulated stochastically.

Relevance to SoftBox

Probably the nearest mechanistic comparator for SoftBox’s central question: shared mechanics changes realized kinetics and displacement.

5. Parallel Cluster Model (2013)

Reduced stochastic motor ensemble



Primary citation

T. Erdmann, P. J. Albert, U. S. Schwarz, J. Chem. Phys. 139, 175104 (2013); arXiv:1307.6510.

Motor representation

N nonprocessive motors in parallel between a common backbone and actin; equivalent states share load equally.

States

Unbound, weakly bound pre-power-stroke, strongly bound post-power-stroke.

Defining laws

One-step master equation for number bound; catch-bond unbinding; power stroke can be blocked by load.

Spatial scale

No explicit binding-site geometry; common extension/load coordinate.

Stochasticity

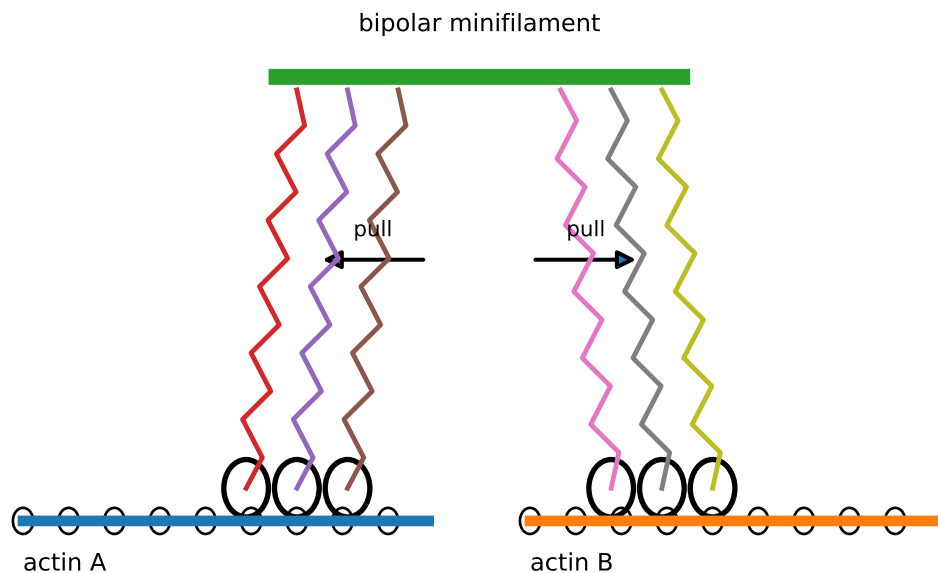
Exact stochastic birth-death dynamics for ensemble occupancy.

Relevance to SoftBox

Captures recruitment, load sharing, and catch-bond stabilization analytically; SoftBox resolves spatial encounters and heterogeneous forces head by head.

6. Bipolar minifilament tug-of-war (2014)

Two coupled motor ensembles



Primary citation

P. J. Albert, T. Erdmann, U. S. Schwarz, New J. Phys. 16, 093019 (2014); arXiv:1404.1587.

Motor representation

Two parallel-cluster ensembles on opposite sides of a bipolar myosin-II minifilament.

States

Each half-filament follows the three-state parallel-cluster cycle.

Defining laws

Two ensemble forces act through the shared minifilament; external elasticity sets force buildup and mechanosensitive attachment.

Spatial scale

Bipolar 1-D crosslink between two actin structures.

Stochasticity

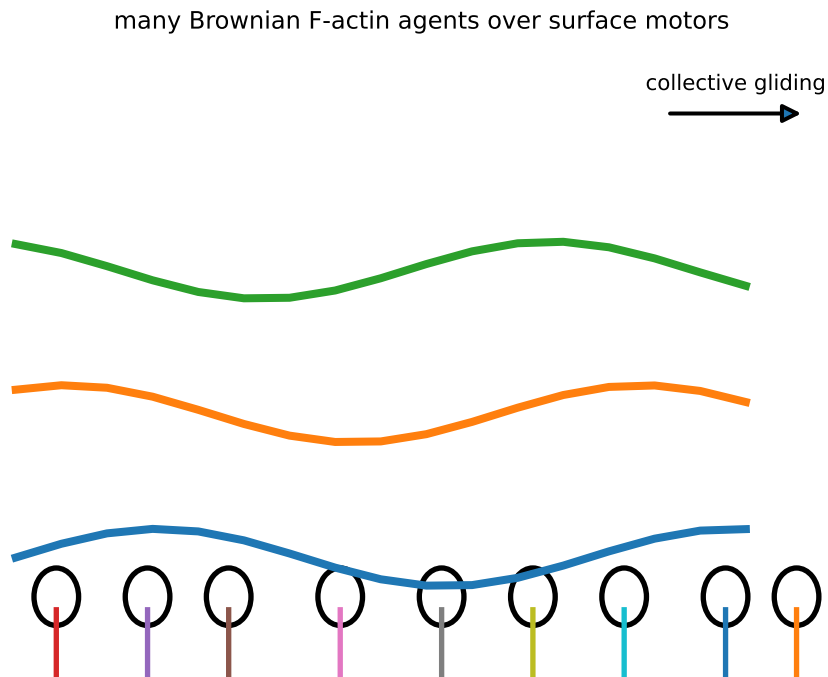
Two coupled stochastic ensemble processes.

Relevance to SoftBox

Architecturally relevant to dimer/minifilament mechanics, but each half is mean-field and lacks explicit head or fork geometry.

7. Jung et al. 2020 motility-assay model (2020)

Agent-based Brownian gliding



Primary citation

W. Jung et al., Soft Matter 16, 1548–1559 (2020), doi:10.1039/C9SM02082A.

Motor representation

Explicit F-actin filaments move on a surface populated by immobile myosin motors; filaments can collide and form collective structures.

States

Motor binding/unbinding and force production are coarse-grained rather than a detailed nucleotide cycle.

Defining laws

Brownian filament mechanics plus motor attachment, elastic force, and prescribed motor action; steric/filament interactions produce collective motion.

Spatial scale

2-D motility assay with multiple flexible filaments.

Stochasticity

Brownian dynamics and stochastic motor events.

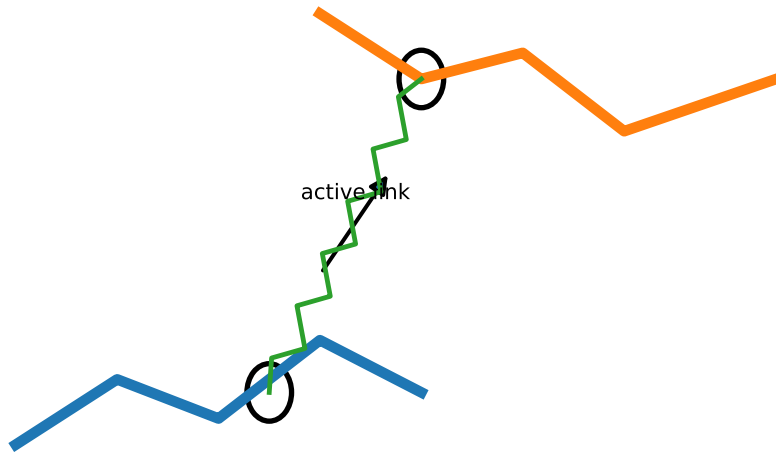
Relevance to SoftBox

Closest in assay scale and simulation style; SoftBox has much finer motor mechanochemistry and internal architecture, but currently a smaller collective-filament scope.

8. Cytosim / Nédélec-Foethke (2009)

Modular cytoskeletal mechanics

modular hands + elastic link + flexible fibers



Primary citation

F. Nédélec and D. Foethke, New J. Phys. 11, 055003 (2009), arXiv:0903.5178.

Motor representation

Flexible fibers are point chains; motors are modular links whose hands bind and move along fibers, often connecting two objects.

States

User-defined binding, unbinding, stepping, end behavior, and oligomerization.

Defining laws

Overdamped Langevin mechanics; constrained fiber length; elastic links; motor hand velocity can depend on force.

Spatial scale

3-D cytoskeletal assemblies from single filaments to cellular structures.

Stochasticity

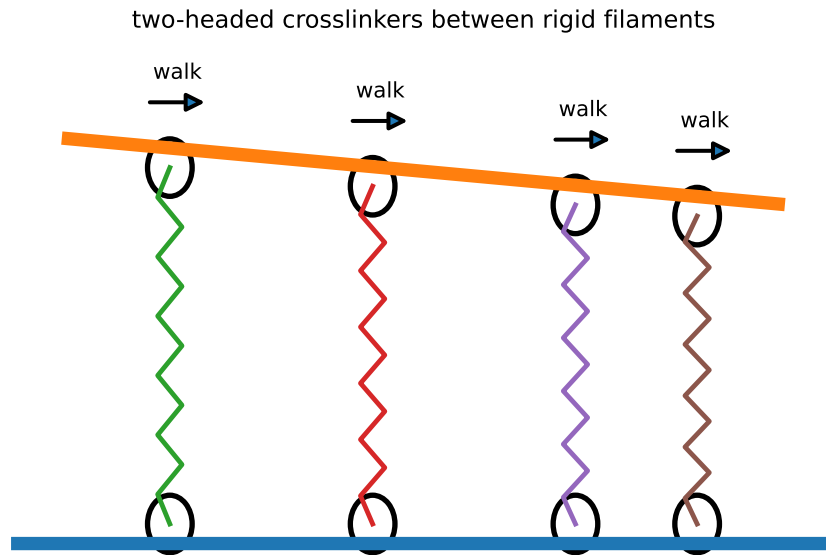
Brownian motion and stochastic binding/unbinding.

Relevance to SoftBox

Much broader spatial scope and modularity; motor internals are typically a spring/link with stepping rules rather than a myosin-specific mechanochemical structure.

9. aLENS / explicit crosslinking motors (2022)

Large-scale active-filament simulation



Primary citation

W. Yan et al., eLife 11:e74160 (2022), doi:10.7554/eLife.74160.

Atlas figure: original simplified reconstruction.

Motor representation

Two-headed crosslinking motors bind pairs of rigid filaments and walk toward filament ends while acting as elastic tethers.

States

Unbound, singly bound, doubly bound; attachment/detachment follow an underlying free-energy landscape.

Defining laws

Motor tether force, force-velocity stepping, thermodynamically consistent binding, rigid-body motion, and hard steric constraints.

Spatial scale

3-D, tens to hundreds of thousands of filaments and motors.

Stochasticity

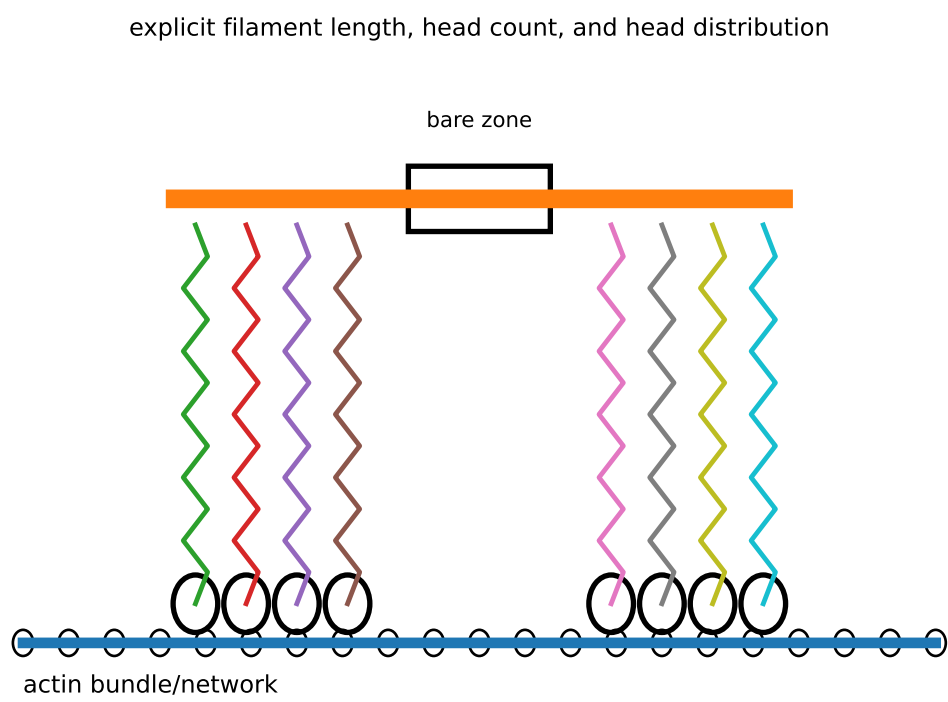
Kinetic Monte Carlo binding plus Brownian/overdamped mechanics.

Relevance to SoftBox

Far larger system scale; motor representation is much coarser. SoftBox can supply a more physical single-motor constitutive model for this class of framework.

10. Ding et al. explicit myosin-II filament architecture (2025)

Agent-based myosin filament / network



Motor representation

Myosin-II filaments have explicit length, bare zone, head number, and head distribution, interacting with actin bundles or networks.

States

Heads bind, unbind, and generate force through a coarse-grained motor rule.

Defining laws

Agent-based filament mechanics; motor heads exert forces determined by attachment geometry and motor kinetics; architecture controls force output.

Spatial scale

Micron-scale bundles and networks with explicit myosin filament geometry.

Stochasticity

Stochastic head interactions and network dynamics.

Relevance to SoftBox

Closest architectural comparator: it resolves where many heads sit on a myosin filament, while SoftBox resolves much more detail inside each head/S2/dimer.

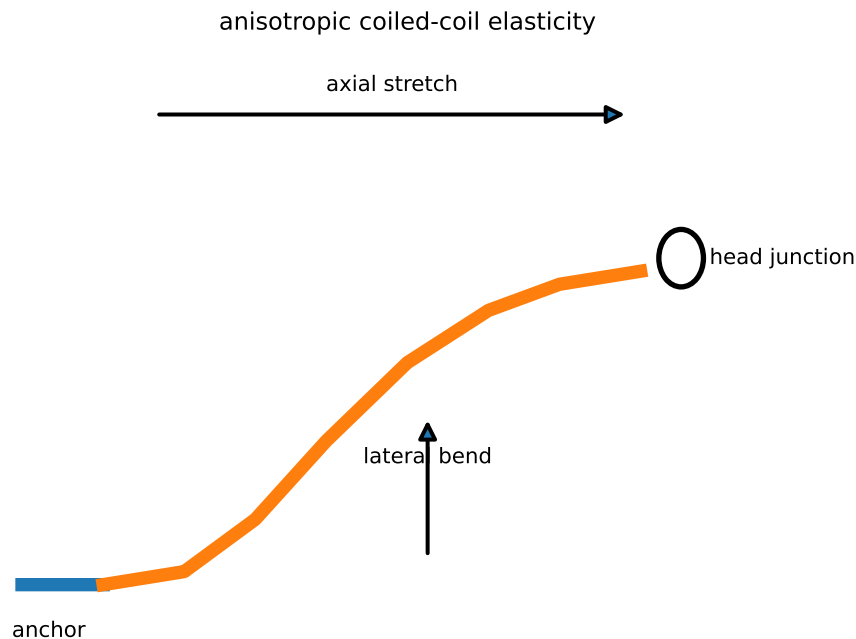
Primary citation

S. Ding et al., eLife 14:e105236 (2025), doi:10.7554/eLife.105236.

Atlas figure: original simplified reconstruction.

11. Adamović-Mijailovich-Karplus S2 mechanics (2008)

Molecular structural mechanics



Primary citation

I. Adamović, S. M. Mijailovich, M. Karplus, Biophys. J. 94, 3779–3789 (2008), doi:10.1529/biophysj.107.122028.

Motor representation

Atomistically characterized myosin-II S2 coiled coil analyzed for axial and lateral elasticity.

States

No motor cycle; structural deformations and normal modes.

Defining laws

Molecular dynamics and normal-mode response yield strongly anisotropic S2 stiffness.

Spatial scale

Nanometer-scale molecular subdomain.

Stochasticity

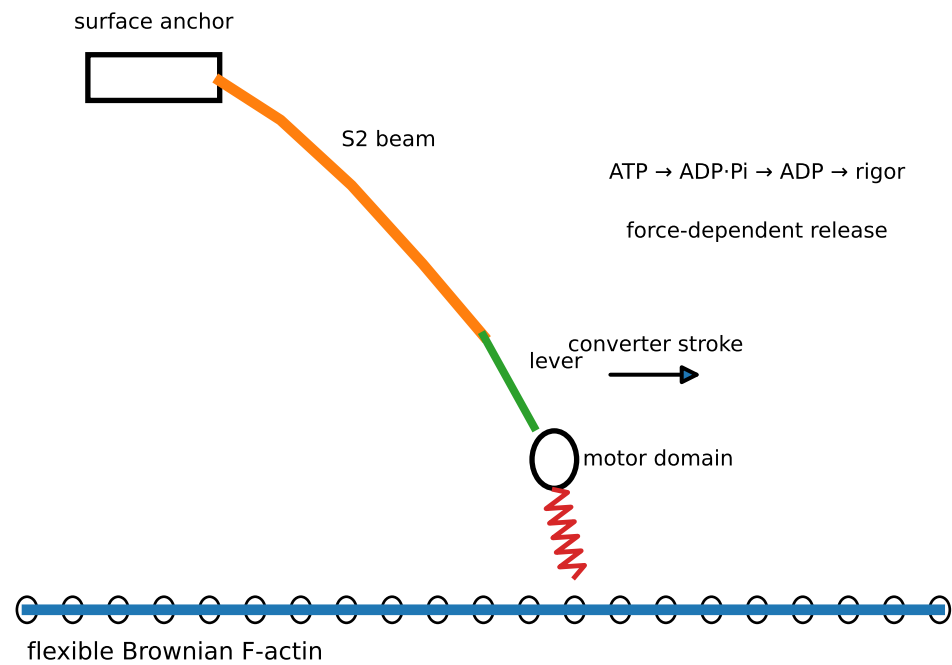
Thermal molecular dynamics.

Relevance to SoftBox

Not a whole motor model, but the closest structural-mechanics source for SoftBox's explicit S2 beam constants and tension-bending asymmetry.

12. SoftBox canonical motor (2026)

Explicit mechanochemical Brownian motor



Primary citation

SoftBox project, canonical motor model v2 (internal model under development).

Motor representation

Explicit motor head/converter/lever, F8 crossbridge, extensible-elastica S2, optional HMM two-head fork, flexible Brownian actin, and surface anchor.

States

Lymn-Taylor nucleotide cycle with force-dependent ADP release and rigor mechanical rupture.

Defining laws

Geometric binding gate; converter stroke; elastic head/S2 mechanics; Brownian filament dynamics; stochastic chemical transitions and rupture.

Spatial scale

3-D motor mechanics coupled to a micron-scale gliding filament and motor lawn.

Stochasticity

Counter-based stochastic chemistry plus Brownian motion; deterministic geometric contact gate.

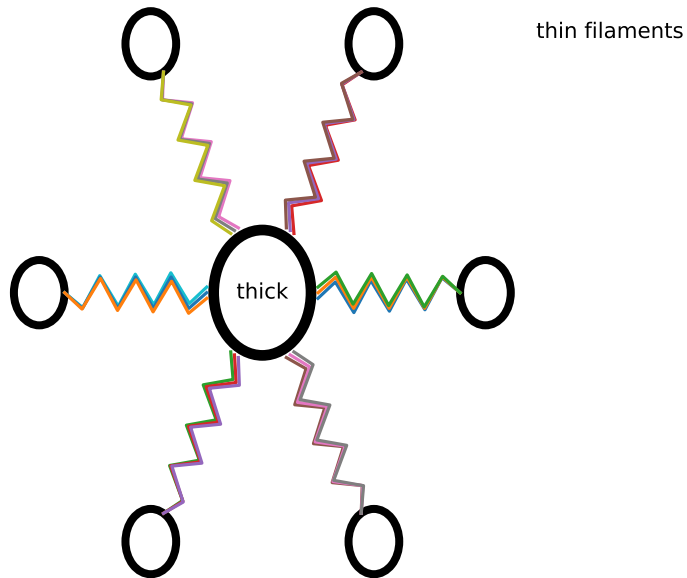
Relevance to SoftBox

Reference model for the atlas: unusually high internal motor detail at an explicit gliding-assay scale.

13. MUSICO 3-D sarcomere lattice (2016)

Spatially explicit multi-filament sarcomere

3-D sarcomere lattice: explicit crowns, target zones, regulation



Primary citation

S. M. Mijailovich et al., “Three-dimensional stochastic model of actin–myosin binding in the sarcomere lattice,” J. Gen. Physiol. 148, 459–488 (2016), doi:10.1085/jgp.201611608.

Atlas figure: original simplified reconstruction.

Motor representation

Explicit hexagonal thick/thin-filament lattice with myosin crowns, actin target zones, regulatory proteins, compliant filaments, and individually tracked cross-bridge events.

States

Multistate actomyosin ATPase cycle plus thin-filament regulatory states; each head can search geometrically permitted actin sites.

Defining laws

3-D geometry determines reach and strain; state-dependent elastic free energies and stochastic transition rates drive attachment, power stroke, and detachment; filament mechanics are solved at ensemble scale.

Spatial scale

Three-dimensional sarcomere lattice with many thick and thin filaments and thousands of potential head-site interactions.

Stochasticity

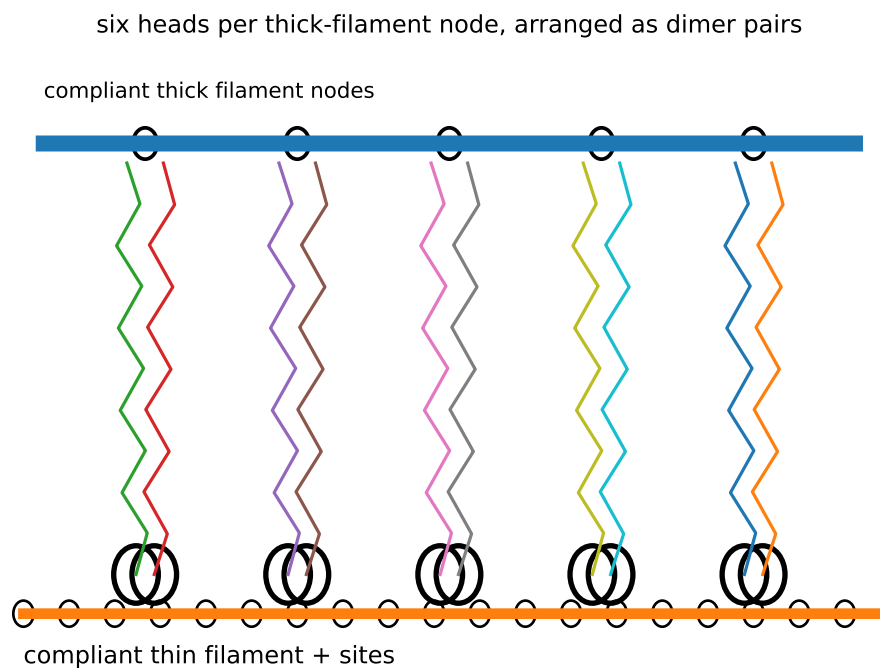
Monte Carlo cross-bridge and regulatory transitions.

Relevance to SoftBox

MUSICO is substantially larger in total system DOF and lattice realism. SoftBox is more mechanically resolved inside one motor (head/lever/F8/S2/fork) and operates in a Brownian gliding assay.

14. FiberSim half-sarcomere (2021)

Explicit-head myofilament simulation



Primary citation

S. Kosta et al., "FiberSim: A flexible open-source model of myofilament-level contraction," J. Gen. Physiol. 153, e202012729 (2021/2022 release), doi:10.1085/jgp.202012729.

Atlas figure: original simplified reconstruction.

Motor representation

Compliant thick and thin filaments discretized into nodes; each thick-filament node carries six myosin heads arranged as three dimer pairs; heads bind nearby active actin sites.

States

Configurable kinetic schemes for myosin and regulatory proteins, including OFF, detached, and attached force-generating states.

Defining laws

Node force balance, filament compliance, head extension-dependent force, nearest-site binding, and user-defined state-transition rate laws.

Spatial scale

Half-sarcomere with explicit axial positions of heads and actin sites.

Stochasticity

Stochastic transitions for individual heads; deterministic mechanical equilibration between events/time steps.

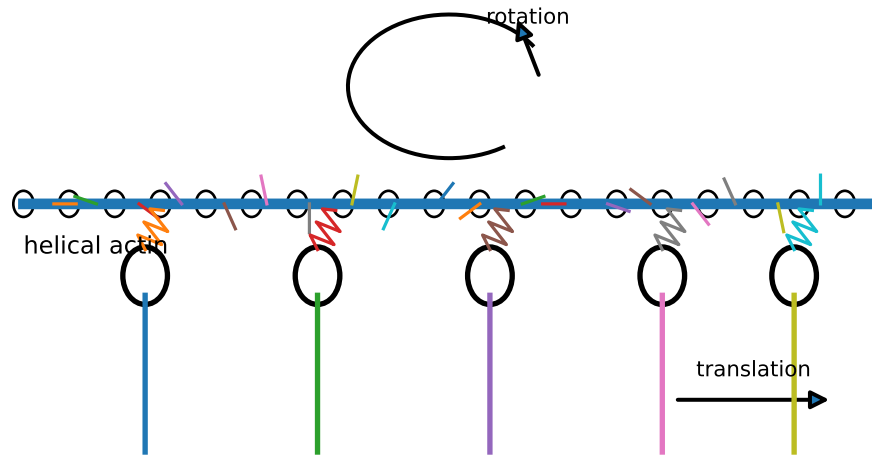
Relevance to SoftBox

FiberSim already has individual heads, dimer pairing, nearest-site geometry, and compliant filaments. SoftBox adds explicit internal head/lever/S2/fork mechanics and free Brownian filament motion.

15. Vilfan 2009 gliding/twirling model (2009)

Helical 3-D gliding geometry

surface-facing target zones bias binding and create torque



Motor representation

Surface-anchored nonprocessive motors interact with the helical actin lattice; accessible target zones depend on monomer azimuth and surface-facing geometry.

States

Coarse attachment, force generation, and detachment cycle.

Defining laws

Stochastic binding to helical target zones plus elastic motor force produces axial translation and torque; filament rotation emerges without an imposed lateral power stroke.

Spatial scale

Gliding filament with axial and rotational coordinates and discrete helical actin sites.

Stochasticity

Stochastic motor attachment and release.

Relevance to SoftBox

A direct predecessor for geometrically explicit motility assays and actin rotation. SoftBox has much richer motor internals and flexible-filament dynamics.

Primary citation

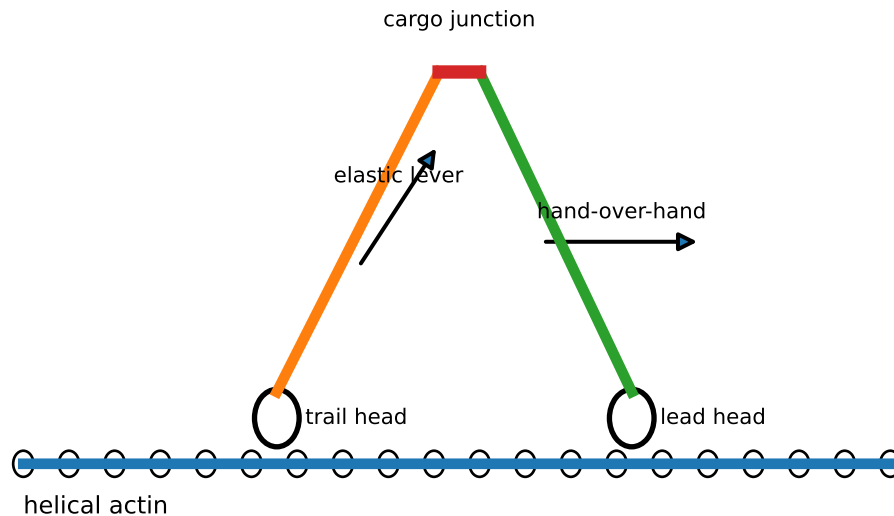
A. Vilfan, "Twirling motion of actin filaments in gliding assays with non-processive myosin motors," *Biophys. J.* 97, 1130-1137 (2009), doi:10.1016/j.bpj.2009.05.059.

Atlas figure: original simplified reconstruction.

16. Vilfan 2005 elastic lever-arm myosin V (2005)

Two-head structural mechanochemistry

two elastic lever arms mechanically coordinate two chemical cycles



Primary citation

A. Vilfan, "Elastic lever-arm model for myosin V," Biophys. J. 88, 3792–3805 (2005), doi:10.1529/biophysj.104.046763.

Atlas figure: original simplified reconstruction.

Motor representation

Two myosin-V heads, each joined to an elastic lever arm; the lever arms meet at a mechanically coupled junction.

States

Four- or five-state nucleotide cycles per head with state-specific lever-arm exit angles.

Defining laws

Elastic-rod bending energy determines accessible two-head configurations; Boltzmann-weighted binding and load-dependent chemical transitions coordinate hand-over-hand stepping.

Spatial scale

Three-dimensional two-head geometry on the helical actin lattice.

Stochasticity

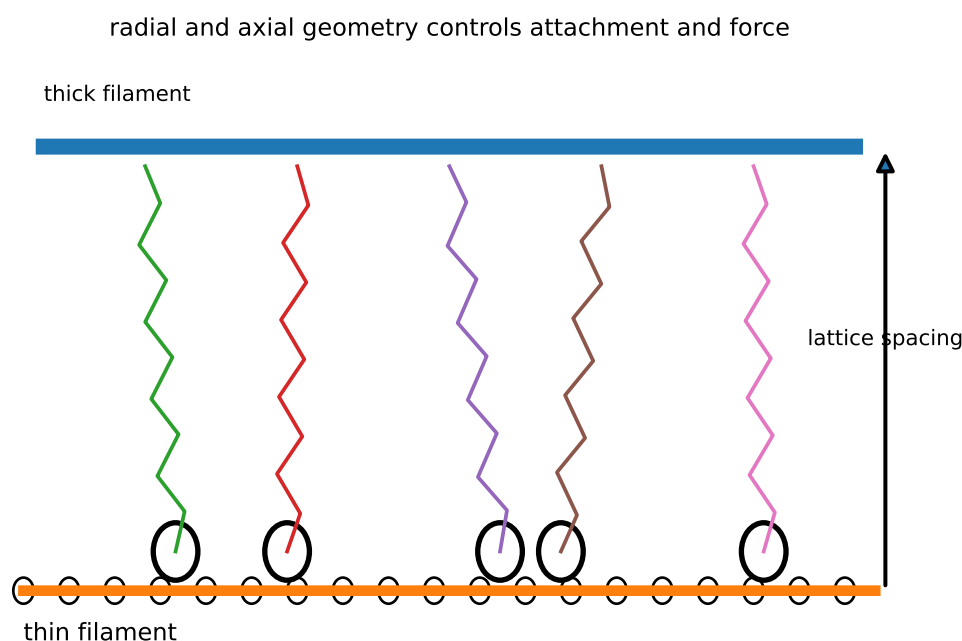
State transitions and site selection are stochastic; geometry and elastic equilibrium determine strain gating.

Relevance to SoftBox

This can rival SoftBox in single-dimer geometric sophistication, but it concerns processive myosin V and uses elastic lever rods rather than explicit myosin-II S2/fork and gliding-law dynamics.

17. Spatially explicit half-sarcomere lattice model (2007-2024)

Geometric cross-bridge lattice



Primary citation

B. C. W. Tanner et al., “Sarcomere lattice geometry influences cooperative myosin binding in muscle,” PLoS Comput. Biol. 3, e115 (2007), doi:10.1371/journal.pcbi.0030115; later spatial-lattice extensions by Powers, Tune, Sponberg and colleagues.

Atlas figure: original simplified reconstruction.

Motor representation

Explicit thick and thin filaments in a lattice; individual cross-bridges attach according to axial and radial separation, with lattice spacing influencing recruitment.

States

Cross-bridge attachment, force-generating, and detachment states coupled to thin-filament activation.

Defining laws

Geometric reach and radial/axial spring strain determine attachment probability and force; filament compliance and lattice spacing feed back on binding.

Spatial scale

Multi-filament half-sarcomere, often with explicit radial lattice coordinates.

Stochasticity

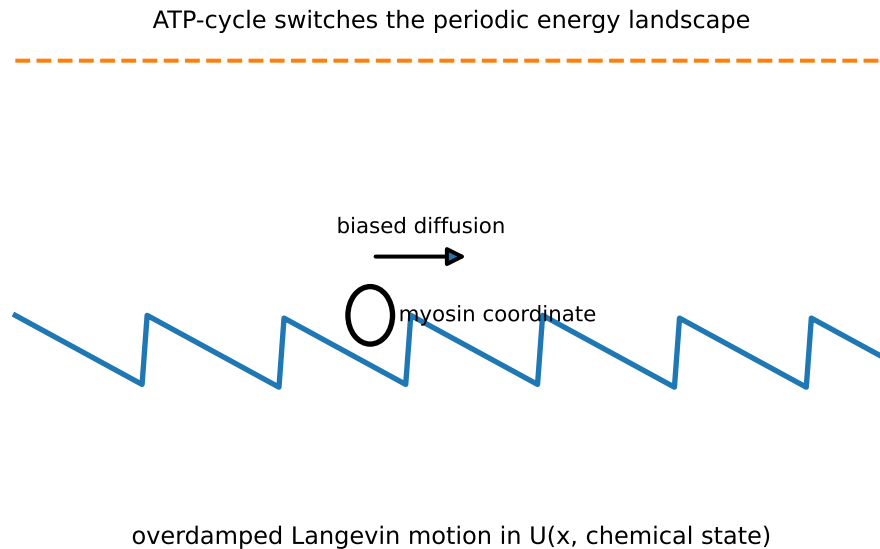
Stochastic cross-bridge cycling and regulatory activation.

Relevance to SoftBox

Strong precedent for spatially explicit geometry controlling recruitment. SoftBox focuses on richer motor internal mechanics and motility rather than sarcomere lattice regulation.

18. Chemically driven fluctuating-ratchet actomyosin (2003)

Energy-landscape Brownian motor



Primary citation

T. Shimokawa, S. Sato, A. Buonocore, L. M. Ricciardi, "A chemically driven fluctuating ratchet model for actomyosin interaction," Phys. Rev. E 69, 051916 (2004), preprint cond-mat/0305074.

Atlas figure: original simplified reconstruction.

Motor representation

A myosin head is a Brownian particle moving in an actin-periodic potential whose shape or state is switched by an external ATP-cycle process.

States

Chemical states select different spatial energy landscapes; the mechanical coordinate diffuses continuously.

Defining laws

Overdamped Langevin motion in a flashing/tilting periodic potential coupled to a chemical Markov process; displacement and energetic cost follow from nonequilibrium stochastic thermodynamics.

Spatial scale

Continuous one-dimensional molecular coordinate along actin.

Stochasticity

Thermal diffusion and stochastic chemical switching are fundamental.

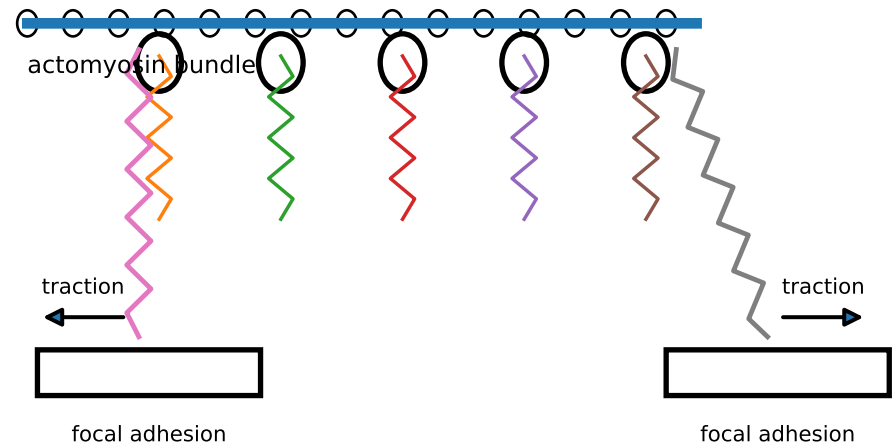
Relevance to SoftBox

More fundamental in energy-landscape thermodynamics, but far less structurally explicit. SoftBox uses explicit mechanical bodies and a biochemical state cycle rather than an imposed periodic potential.

19. Cell-traction multiscale individual-head model (2022)

Motor-to-adhesion multiscale model

individual motor impulses feed bundle and substrate mechanics



Primary citation

E. Vázquez-Hidalgo et al., “Chemo-mechanical factors that limit cellular force generation,” *Front. Phys.* 10, 831776 (2022), doi:10.3389/fphy.2022.831776.

Atlas figure: original simplified reconstruction.

Motor representation

Individual myosin motor-head states and actomyosin units are coupled upward to adhesion/substrate mechanics to estimate cellular traction.

States

Nanoscale motor-state cycle embedded in filament and adhesion-scale dynamics.

Defining laws

Motor kinetics and force impulses feed actomyosin bundle mechanics and focal-adhesion/substrate force transmission across multiple time scales.

Spatial scale

From nanometer motor events to micron-scale adhesion regions.

Stochasticity

Individual motor-state activity is stochastic or probabilistic; larger-scale force transmission is coarse-grained.

Relevance to SoftBox

Broader multiscale output but a reduced individual-motor geometry. It is an important warning against claiming that SoftBox alone bridges nanoscale motor behavior to micron-scale force.